

SECTION INDEX: Cotter & Knuckle Joints Classification Index

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Socket and Spigot Cotter Joint	Design 30 kN joint ($d, d_2, t, d_1, b, d_4, c, a, d_3, t_1$), crushing & shear checks	Section 1
Sleeve and Cotter Joint	Design 60 kN sleeve joint (d, d_2, t, d_1, b, a, c), double shear & crushing checks	Section 2
Gib and Cotter Joint	Design 50 kN connecting rod big end joint ($B_1, t, t_1, t_3, B, b_1, b, t_4, l_1, l_2$)	Section 3
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Design Standard Note: All design equations, shear failures, crushing stress limits, and empirical proportion checks conform strictly to standard mechanical design principles (R.S. Khurmi & J.K. Gupta).

SECTION 1: Socket and Spigot Cotter Joint (30 kN Load)

System Parameters

Tension / Compression Load: $P = 30 \text{ kN} = 30000 \text{ N}$

Carbon Steel Stresses: Tensile/Compressive $\sigma_t = \sigma_c = 50 \text{ MPa}$, Shear $\tau = 35 \text{ MPa}$, Crushing $\sigma_c = 90 \text{ MPa}$

Design Protocol

Step 1: Size rod diameter d in tension

Step 2: Size spigot diameter d_2 & cotter thickness t (Trial & Crushing Check)

Step 3: Size socket outer diameter d_1 , cotter width b , socket collar d_4, c , rod end a , & spigot collar d_3, t_1

1. Rod Diameter (d)

Tensile failure of rod

$$P = \frac{\pi}{4} d^2 \cdot \sigma_t$$

$$30000 = \frac{\pi}{4} d^2 (50) = 39.3 d^2$$

$$d^2 = \frac{30000}{39.3} = 763$$

$$d = 27.6 \text{ mm}$$

$$\Rightarrow d = 28 \text{ mm} \quad (\text{Ans.})$$

2. Trial Spigot Diameter (d_2) & Cotter Thickness (t)

Tension failure of spigot across slot ($t = \frac{d_2}{4}$)

$$P = \left[\frac{\pi}{4} d_2^2 - d_2 t \right] \sigma_t$$

$$30000 = \left[\frac{\pi}{4} d_2^2 - d_2 \left(\frac{d_2}{4} \right) \right] 50 = 26.8 (d_2)^2$$

$$(d_2)^2 = \frac{30000}{26.8} = 1119.4$$

$$d_2 = 33.4 \text{ mm} \Rightarrow 34 \text{ mm}, \quad t = \frac{34}{4} = 8.5 \text{ mm}$$

3. Crushing Stress Check & Recalculation (d_2, t)

Check crushing stress for trial dimensions

$$\sigma_c = \frac{P}{d_2 \cdot t} = \frac{30000}{34 \times 8.5} = 103.8 \text{ N/mm}^2 > 90 \text{ MPa} \quad (\text{UNSAFE})$$

$$30000 = d_2 \left(\frac{d_2}{4} \right) 90 = 22.5 (d_2)^2$$

$$(d_2)^2 = \frac{30000}{22.5} = 1333$$

$$d_2 = 40 \text{ mm} \quad (\text{Ans.})$$

$$t = \frac{40}{4} = 10 \text{ mm} \quad (\text{Ans.})$$

4. Socket Outer Diameter (d_1)

Tension failure of socket across slot

$$P = \left[\frac{\pi}{4} (d_1^2 - d_2^2) - (d_1 - d_2)t \right] \sigma_t$$

$$30000 = \left[\frac{\pi}{4} (d_1^2 - 40^2) - (d_1 - 40)(10) \right] 50$$

$$0.7854d_1^2 - 10d_1 - 1854.6 = 0$$

$$d_1 = 49.9 \text{ mm}$$

$$\Rightarrow d_1 = \mathbf{50 \text{ mm}} \quad (\text{Ans.})$$

5. Width of Cotter (b)

Double shear failure of cotter

$$P = 2b \cdot t \cdot \tau$$

$$30000 = 2b \times 10 \times 35 = 700b$$

$$b = \frac{30000}{700} = \mathbf{43 \text{ mm}} \quad (\text{Ans.})$$

6. Socket Collar Diameter (d_4)

Crushing failure of socket collar

$$P = (d_4 - d_2)t \cdot \sigma_c$$

$$30000 = (d_4 - 40) \times 10 \times 90 = 900(d_4 - 40)$$

$$d_4 - 40 = \frac{30000}{900} = 33.3$$

$$d_4 = \mathbf{73.3 \text{ mm}}$$

$$\Rightarrow d_4 = \mathbf{75 \text{ mm}} \quad (\text{Ans.})$$

7. Socket Collar Thickness (c)

Double shear failure of socket end

$$P = 2(d_4 - d_2)c \cdot \tau$$

$$30000 = 2(75 - 40)c \times 35 = 2450c$$

$$c = \frac{30000}{2450} = 12 \text{ mm (Ans.)}$$

8. Slot End to Rod End Distance (a)

Double shear failure of rod end

$$P = 2a \cdot d_2 \cdot \tau$$

$$30000 = 2a \times 40 \times 35 = 2800a$$

$$a = \frac{30000}{2800} = 10.7 \text{ mm}$$

$$\Rightarrow a = 11 \text{ mm (Ans.)}$$

9. Spigot Collar Diameter (d_3)

Crushing failure of spigot collar

$$P = \frac{\pi}{4}(d_3^2 - d_2^2)\sigma_c$$

$$30000 = \frac{\pi}{4}(d_3^2 - 40^2)90$$

$$d_3^2 - 1600 = 424$$

$$d_3 = \sqrt{2024} = 45 \text{ mm (Ans.)}$$

10. Spigot Collar Thickness (t_1)

Shear failure of spigot collar

$$P = \pi d_2 \cdot t_1 \cdot \tau$$

$$30000 = \pi \times 40 \times t_1 \times 35 = 4400t_1$$

$$t_1 = \frac{30000}{4400} = 6.8 \text{ mm}$$

$$\Rightarrow t_1 = 8 \text{ mm} \quad (\text{Ans.})$$

11. Cotter Length (l) & Clearance Dimension (e)

Empirical length and clearance

$$l = 4d = 4 \times 28 = 112 \text{ mm} \quad (\text{Ans.})$$

$$e = 1.2d = 1.2 \times 28 = 33.6 \text{ mm}$$

$$\Rightarrow e = 34 \text{ mm} \quad (\text{Ans.})$$

12. Complete Design Output

Socket & Spigot Joint Summary: $d = 28 \text{ mm}$, $d_2 = 40 \text{ mm}$, $t = 10 \text{ mm}$, $d_1 = 50 \text{ mm}$, $b = 43 \text{ mm}$, $d_4 = 75 \text{ mm}$, $c = 12 \text{ mm}$, $a = 11 \text{ mm}$, $d_3 = 45 \text{ mm}$, $t_1 = 8 \text{ mm}$, $l = 112 \text{ mm}$, $e = 34 \text{ mm}$

SECTION 1: Socket and Spigot Cotter Joint Assembly — MECHANICAL DIAGRAM

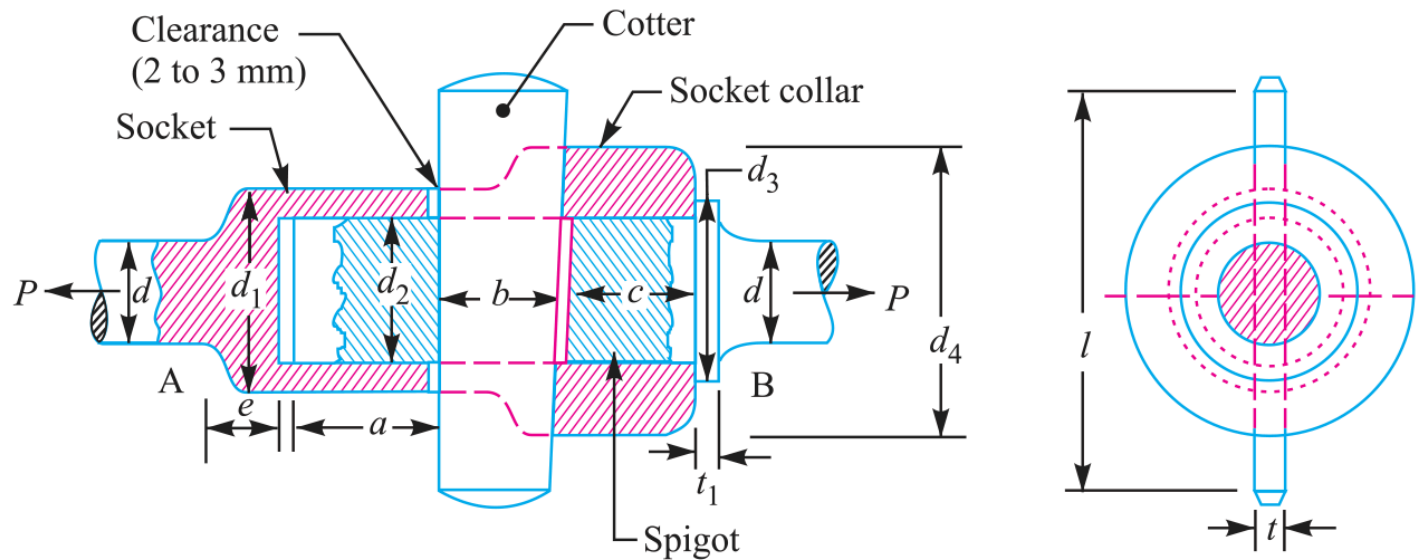


Fig. 12.1. Socket and spigot cotter joint.

Figure 12.1: Socket & Spigot Cotter Joint Assembly & Proportions

SECTION 2: Sleeve and Cotter Joint (60 kN Load)

System Parameters

Tensile Load: $P = 60 \text{ kN} = 60000 \text{ N}$

Allowable Stresses: Tensile $\sigma_t = 60 \text{ MPa}$, Shear $\tau = 70 \text{ MPa}$, Crushing $\sigma_c = 125 \text{ MPa}$

Design Protocol

Step 1: Size rod diameter d in tension

Step 2: Size enlarged rod end d_2 & cotter thickness t ($t = \frac{d_2}{4}$) with crushing check

Step 3: Size sleeve outer diameter d_1 , cotter width b , rod shear distance a , & sleeve shear distance c

1. Rod Diameter (d)

Tensile failure of rod

$$P = \frac{\pi}{4} d^2 \cdot \sigma_t$$

$$60000 = \frac{\pi}{4} d^2 (60) = 47.13 d^2$$

$$d^2 = \frac{60000}{47.13} = 1273$$

$$d = 35.7 \text{ mm}$$

$$\Rightarrow d = 36 \text{ mm} \quad (\text{Ans.})$$

2. Enlarged Rod Diameter (d_2) & Cotter Thickness (t)

Tension failure of rod across slot ($t = \frac{d_2}{4}$)

$$P = \left[\frac{\pi}{4} d_2^2 - d_2 t \right] \sigma_t$$

$$60000 = \left[\frac{\pi}{4} d_2^2 - d_2 \left(\frac{d_2}{4} \right) \right] 60 = 32.13 (d_2)^2$$

$$(d_2)^2 = \frac{60000}{32.13} = 1867$$

$$d_2 = 43.2 \text{ mm}$$

$$\Rightarrow d_2 = 44 \text{ mm} \quad (\text{Ans.})$$

$$t = \frac{44}{4} = 11 \text{ mm} \quad (\text{Ans.})$$

3. Crushing Stress Check

Check crushing stress on rod end / cotter

$$\sigma_c = \frac{P}{d_2 \cdot t}$$

$$= \frac{60000}{44 \times 11}$$

$$= 124 \text{ N/mm}^2 \leq 125 \text{ MPa} \quad (\text{SAFE})$$

4. Sleeve Outer Diameter (d_1)

Tension failure of sleeve across slot

$$P = \left[\frac{\pi}{4} (d_1^2 - d_2^2) - (d_1 - d_2)t \right] \sigma_t$$

$$60000 = \left[\frac{\pi}{4} (d_1^2 - 44^2) - (d_1 - 44)(11) \right] 60$$

$$0.7854d_1^2 - 11d_1 - 1036.7 = 0$$

$$d_1 = 58.4 \text{ mm}$$

$$\Rightarrow d_1 = 60 \text{ mm} \quad (\text{Ans.})$$

5. Width of Cotter (b)

Double shear failure of cotter

$$P = 2b \cdot t \cdot \tau$$

$$60000 = 2b \times 11 \times 70 = 1540b$$

$$b = \frac{60000}{1540} = 38.96 \text{ mm}$$

$$\Rightarrow b = 40 \text{ mm} \quad (\text{Ans.})$$

6. Rod End Shear Distance (a)

Double shear failure of rod end

$$P = 2a \cdot d_2 \cdot \tau$$

$$60000 = 2a \times 44 \times 70 = 6160a$$

$$a = \frac{60000}{6160} = 9.74 \text{ mm}$$

$$\Rightarrow a = 10 \text{ mm} \quad (\text{Ans.})$$

7. Sleeve End Shear Distance (c)

Double shear failure of sleeve end

$$P = 2(d_1 - d_2)c \cdot \tau$$

$$60000 = 2(60 - 44)c \times 70 = 2240c$$

$$c = 26.78 \text{ mm}$$

$$\Rightarrow c = 28 \text{ mm} \quad (\text{Ans.})$$

8. Complete Design Output

Sleeve & Cotter Joint Summary: $d = 36 \text{ mm}$, $d_2 = 44 \text{ mm}$, $t = 11 \text{ mm}$, $d_1 = 60 \text{ mm}$, $b = 40 \text{ mm}$, $a = 10 \text{ mm}$, $c = 28 \text{ mm}$

SECTION 2: Sleeve and Cotter Joint Assembly — MECHANICAL DIAGRAM

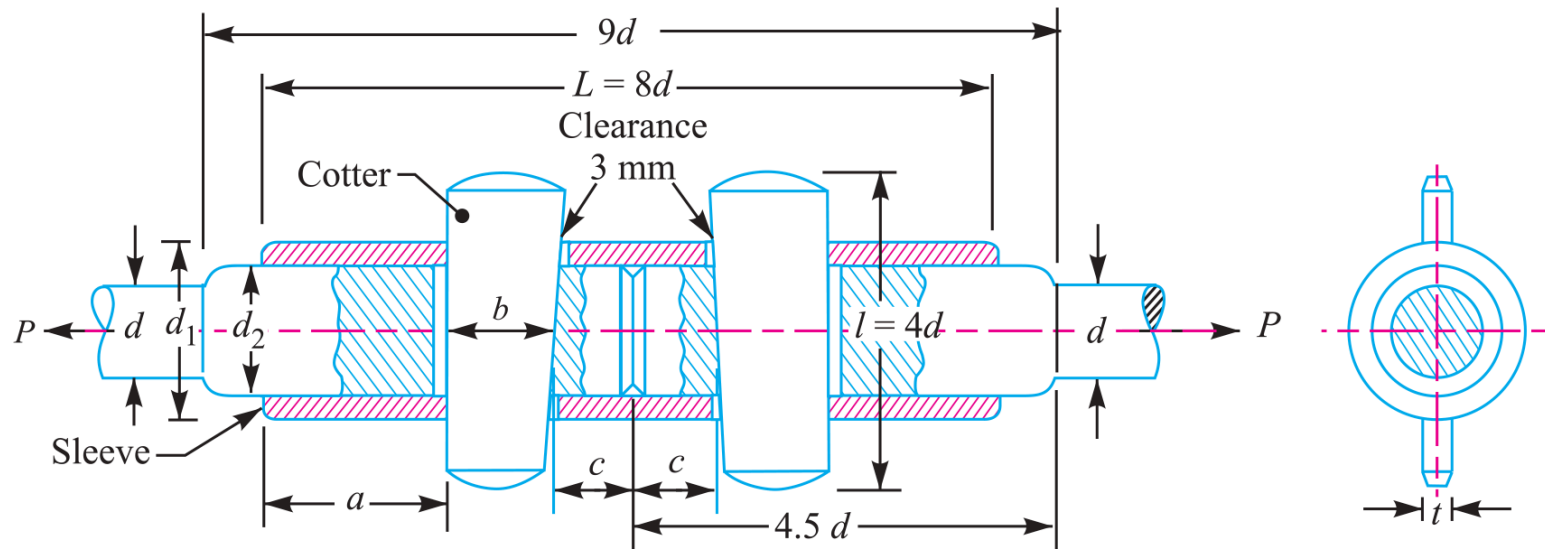


Fig. 12.9. Sleeve and cotter joint.

Figure 12.9: Sleeve & Cotter Joint Assembly & Proportions

SECTION 3: Gib and Cotter Joint for Connecting Rod Big End (50 kN Load)

System Parameters

Max Load: $P = 50 \text{ kN} = 50000 \text{ N}$ | Rod Diameter Adjacent to Strap: $d = 75 \text{ mm}$

Strap Tensile Stress: $\sigma_t = 25 \text{ MPa}$ | Cotter & Gib Shear Stress: $\tau = 20 \text{ MPa}$

Design Protocol

Step 1: Determine strap width $B_1 = d = 75 \text{ mm}$, cotter/gib thickness $t = \frac{B_1}{4} = 20 \text{ mm}$

Step 2: Size strap thinnest section t_1 , strap thickness at cotter t_3 , & total width B (divided into gib b_1 & cotter b)

Step 3: Determine strap end t_4 , rod end l_1 , & overhang l_2

1. Strap Width (B_1) & Cotter/Gib Thickness (t)

Width equal to rod diameter $d = 75 \text{ mm}$

$$B_1 = d = 75 \text{ mm} \quad (\text{Ans.})$$

$$t = \frac{B_1}{4} = \frac{75}{4} = 18.75 \text{ mm}$$

$$\Rightarrow t = 20 \text{ mm} \quad (\text{Ans.})$$

2. Thinnest Strap Thickness (t_1)

Tensile failure of strap at thinnest section

$$P = 2B_1 \cdot t_1 \cdot \sigma_t$$

$$50000 = 2 \times 75 \times t_1 \times 25 = 3750t_1$$

$$t_1 = \frac{50000}{3750} = 13.3 \text{ mm}$$

$$\Rightarrow t_1 = 15 \text{ mm} \quad (\text{Ans.})$$

3. Strap Thickness at Cotter Hole (t_3)

Equal cross-sectional area criterion

$$2t_3(B_1 - t) = 2t_1B_1$$

$$2t_3(75 - 20) = 2 \times 15 \times 75$$

$$110t_3 = 2250$$

$$t_3 = \frac{2250}{110} = 20.45 \text{ mm}$$

$$\Rightarrow t_3 = 21 \text{ mm} \quad (\text{Ans.})$$

4. Total Width of Gib & Cotter (B)

Double shear failure of gib and cotter

$$P = 2B \cdot t \cdot \tau$$

$$50000 = 2B \times 20 \times 20 = 800B$$

$$B = \frac{50000}{800} = 62.5 \text{ mm}$$

$$\Rightarrow B = 65 \text{ mm} \quad (\text{Ans.})$$

5. Division of Width between Gib (b_1) & Cotter (b)

Empirical proportions (55% gib, 45% cotter)

$$b_1 = 0.55B = 0.55 \times 65 = 35.75 \text{ mm}$$

$$\Rightarrow b_1 = 36 \text{ mm} \quad (\text{Ans.})$$

$$b = 0.45B = 0.45 \times 65 = 29.25 \text{ mm}$$

$$\Rightarrow b = 30 \text{ mm} \quad (\text{Ans.})$$

6. Empirical Proportions (t_4, l_1, l_2)

Strap end, rod end, and overhang dimensions

$$t_4 = 1.25t_1 = 1.25 \times 15 = 18.75 \text{ mm}$$

$$\Rightarrow t_4 = 20 \text{ mm} \quad (\text{Ans.})$$

$$l_1 = 2t_1 = 2 \times 15 = 30 \text{ mm} \quad (\text{Ans.})$$

$$l_2 = 2.5t_1 = 2.5 \times 15 = 37.5 \text{ mm}$$

$$\Rightarrow l_2 = 40 \text{ mm} \quad (\text{Ans.})$$

7. Complete Design Output

Gib & Cotter Joint Summary: $B_1 = 75 \text{ mm}$, $t = 20 \text{ mm}$, $t_1 = 15 \text{ mm}$, $t_3 = 21 \text{ mm}$, $B = 65 \text{ mm}$ (Gib $b_1 = 36 \text{ mm}$, Cotter $b = 30 \text{ mm}$), $t_4 = 20 \text{ mm}$, $l_1 = 30 \text{ mm}$, $l_2 = 40 \text{ mm}$

SECTION 3: Gib and Cotter Joint Assembly — MECHANICAL DIAGRAM

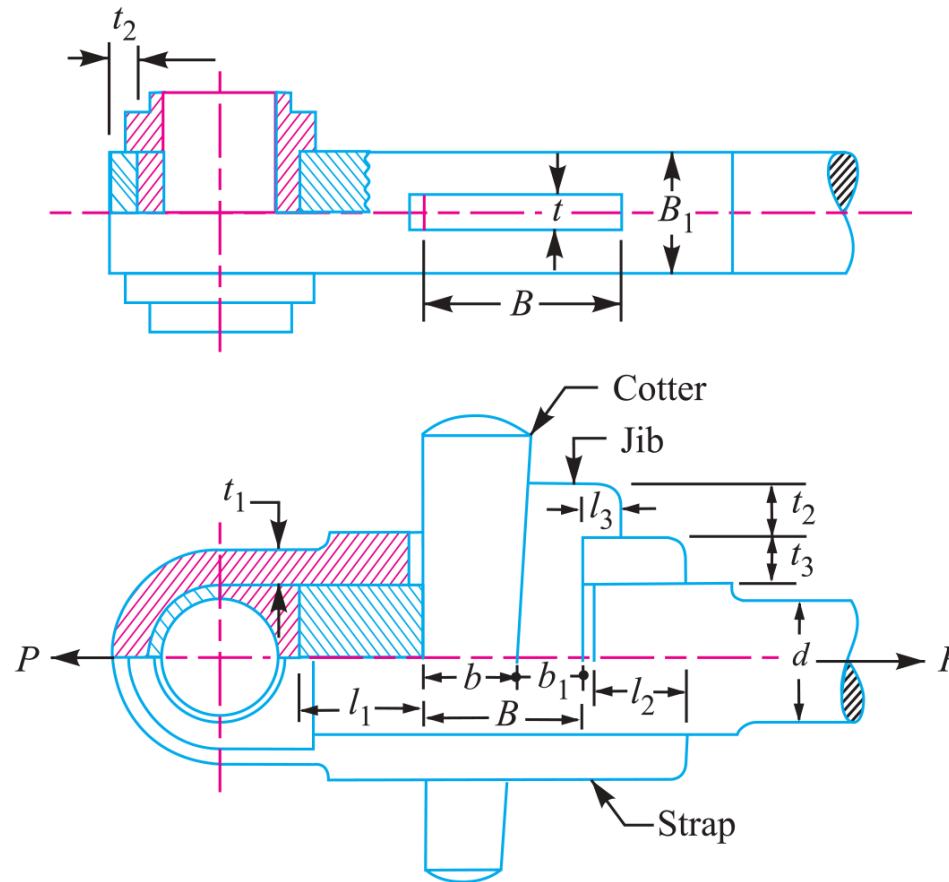


Fig. 12.12. Gib and cotter joint for strap end of a connecting rod.

Figure 12.12: Gib and Cotter Joint Construction for Connecting Rod Big End

SECTION 4: Knuckle Joint Design & Stress Checks (150 kN Load)

System Parameters

Tensile Load: $P = 150 \text{ kN} = 150000 \text{ N}$

Allowable Stresses: Tension $\sigma_t = 75 \text{ MPa}$, Shear $\tau = 60 \text{ MPa}$, Compression $\sigma_c = 150 \text{ MPa}$

Design Protocol

Step 1: Size rod diameter d in tension & calculate empirical dimensions ($d_1, d_2, d_3, t, t_1, t_2$)

Step 2: Verify pin shear, single eye tension/shear/crushing, & fork end tension/shear/crushing

1. Rod Diameter (d) & Empirical Proportions

Tensile failure of solid rod

$$P = \frac{\pi}{4} d^2 \cdot \sigma_t$$

$$150000 = \frac{\pi}{4} d^2 (75) = 59d^2$$

$$d^2 = \frac{150000}{59} = 2540$$

$$d = 52 \text{ mm} \quad (\text{Ans.})$$

$$d_1 = d = 52 \text{ mm}, \quad d_2 = 2d = 104 \text{ mm}$$

$$d_3 = 1.5d = 78 \text{ mm}, \quad t = 1.25d = 65 \text{ mm}$$

$$t_1 = 0.75d = 40 \text{ mm}, \quad t_2 = 0.5d = 26 \text{ mm}$$

2. Knuckle Pin Double Shear Check (τ)

Shear stress check on pin

$$P = 2\left(\frac{\pi}{4}d_1^2\right)\tau$$

$$150000 = 2\left(\frac{\pi}{4}\right)(52)^2\tau = 4248\tau$$

$$\tau = \frac{150000}{4248} = 35.3 \text{ MPa} \leq 60 \text{ MPa} \quad (\text{SAFE})$$

3. Single Eye Tension Check (σ_t)

Tearing across pin hole

$$P = (d_2 - d_1)t \cdot \sigma_t$$

$$150000 = (104 - 52)65 \cdot \sigma_t = 3380\sigma_t$$

$$\sigma_t = \frac{150000}{3380} = 44.4 \text{ MPa} \leq 75 \text{ MPa} \quad (\text{SAFE})$$

4. Single Eye Shear Check (τ)

Double shearing of eye end

$$P = (d_2 - d_1)t \cdot \tau$$

$$150000 = (104 - 52)65 \cdot \tau = 3380\tau$$

$$\tau = \frac{150000}{3380} = 44.4 \text{ MPa} \leq 60 \text{ MPa} \quad (\text{SAFE})$$

5. Single Eye Crushing Check (σ_c)

Crushing between pin and eye

$$P = d_1 \cdot t \cdot \sigma_c$$

$$150000 = 52 \times 65 \cdot \sigma_c = 3380\sigma_c$$

$$\sigma_c = \frac{150000}{3380} = 44.4 \text{ MPa} \leq 150 \text{ MPa} \quad (\text{SAFE})$$

6. Fork End Tension Check (σ_t)

Tearing across fork pin holes

$$P = (d_2 - d_1)2t_1 \cdot \sigma_t$$

$$150000 = (104 - 52)2(40) \cdot \sigma_t = 4160\sigma_t$$

$$\sigma_t = \frac{150000}{4160} = 36 \text{ MPa} \leq 75 \text{ MPa} \quad (\text{SAFE})$$

7. Fork End Shear Check (τ)

Double shearing of fork ends

$$P = (d_2 - d_1)2t_1 \cdot \tau$$

$$150000 = (104 - 52)2(40) \cdot \tau = 4160\tau$$

$$\tau = \frac{150000}{4160} = 36 \text{ MPa} \leq 60 \text{ MPa} \quad (\text{SAFE})$$

8. Fork End Crushing Check (σ_c)

Crushing between pin and fork

$$P = d_1 \cdot 2t_1 \cdot \sigma_c$$

$$150000 = 52 \times 2(40) \cdot \sigma_c = 4160\sigma_c$$

$$\sigma_c = \frac{150000}{4160} = 36 \text{ MPa} \leq 150 \text{ MPa} \quad (\text{SAFE})$$

9. Complete Design Output

Knuckle Joint Summary: $d = 52$ mm, $d_1 = 52$ mm, $d_2 = 104$ mm, $d_3 = 78$ mm, $t = 65$ mm, $t_1 = 40$ mm, $t_2 = 26$ mm, All 8 Stress Checks SAFE

Figure 12.16: Knuckle Joint Construction & Pin Assembly Details

SECTION 5: Knuckle Joint Design with Factor of Safety (70 kN Load)

System Parameters

Pull Load: $P = 70 \text{ kN} = 70000 \text{ N}$ | Factor of Safety: $\text{FOS} = 6$

Rod Ultimate Strength: $\sigma_{tu} = 420 \text{ MPa} \Rightarrow \text{Permissible } \sigma_t = \frac{420}{6} = 70 \text{ N/mm}^2$

Pin Ultimate Shear Strength: $\tau_u = 396 \text{ MPa} \Rightarrow \text{Permissible } \tau = \frac{396}{6} = 66 \text{ N/mm}^2$

Design Protocol

Step 1: Size rod diameter d in tension & calculate empirical dimensions (d_1, d_2, d_3, t, t_1)

Step 2: Verify pin shear stress τ , single eye tension σ_t , & fork end tension σ_t

1. Rod Diameter (d) & Empirical Proportions

Tensile failure of tie rod under allowable $\sigma_t = 70 \text{ MPa}$

$$P = \frac{\pi}{4} d^2 \cdot \sigma_t$$

$$70000 = \frac{\pi}{4} d^2 (70) = 55d^2$$

$$d^2 = \frac{70000}{55} = 1273$$

$$d = 36 \text{ mm} \quad (\text{Ans.})$$

$$d_1 = d = 36 \text{ mm}, \quad d_2 = 2d = 72 \text{ mm}$$

$$d_3 = 1.5d = 54 \text{ mm}, \quad t = 1.25d = 45 \text{ mm}$$

$$t_1 = 0.75d = 27 \text{ mm}$$

2. Knuckle Pin Double Shear Check (τ)Check shear stress for allowable $\tau = 66$ MPa

$$P = 2\left(\frac{\pi}{4}d_1^2\right)\tau$$

$$70000 = 2\left(\frac{\pi}{4}\right)(36)^2\tau = 2036\tau$$

$$\tau = \frac{70000}{2036} = 34.4 \text{ MPa} \leq 66 \text{ MPa} \quad (\text{SAFE})$$

3. Single Eye Tension Check (σ_t)Check tensile stress for allowable $\sigma_t = 70$ MPa

$$P = (d_2 - d_1)t \cdot \sigma_t$$

$$70000 = (72 - 36)45 \cdot \sigma_t = 1620\sigma_t$$

$$\sigma_t = \frac{70000}{1620} = 43.2 \text{ MPa} \leq 70 \text{ MPa} \quad (\text{SAFE})$$

4. Fork End Tension Check (σ_t)Check tensile stress for allowable $\sigma_t = 70$ MPa

$$P = (d_2 - d_1)2t_1 \cdot \sigma_t$$

$$70000 = (72 - 36)2(27) \cdot \sigma_t = 1944\sigma_t$$

$$\sigma_t = \frac{70000}{1944} = 36 \text{ MPa} \leq 70 \text{ MPa} \quad (\text{SAFE})$$

5. Complete Design Output

Knuckle Joint (FOS = 6) Summary: d = 36 mm, d1 = 36 mm, d2 = 72 mm, d3 = 54 mm, t = 45 mm, t1 = 27 mm, All Stress Checks SAFE

Figure 12.16: Knuckle Joint Fork & Pin Details